Structural Diagnosability and Phase Logic

Abstract

This paper addresses the problem of *structural diagnosability* of enterprise states under conditions of incomplete and indirect observability. Observable metrics, indicators, and outcome variables generally fail to uniquely determine the internal structural configurations that generate them. As a result, distinct enterprise structures may induce indistinguishable observable projections, imposing intrinsic limits on diagnostic inference.

Within the formal framework of the Enterprise Continuum K_{EA} and its executable specification ETS.K_EA.v1.1, structural diagnosability is analyzed as an epistemic property of the mapping from internal structural spaces to observable domains. The analysis is strictly diagnostic and non-prescriptive: it does not address control, optimization, intervention, or corrective action. Its purpose is to characterize when an enterprise state can be inferred to belong to a *structural phase class*, rather than identified as a unique configuration.

Three contributions are developed. First, structural diagnosability is formally defined in terms of distinguishability between equivalence classes induced by observable projections. Second, diagnostic claims are interpreted within a phase-based framework, establishing that, in general, well-founded diagnoses are phase-level rather than state-level assertions. Third, falsifiability conditions for diagnostic statements are specified by identifying observable distinctions whose occurrence would refute phase membership.

The scope of the results is intentionally limited. No claims are made regarding desirable states, optimal structures, or recommended actions. Non-diagnosability is treated as a valid and informative outcome, reflecting fundamental epistemic constraints rather than deficiencies of the model. Under these conditions, structural diagnosability is established as a formally bounded property of enterprise models, jointly determined by observability and structural organization.

1 Problem Statement

Enterprises are complex systems whose internal structure is only partially accessible to observation. Architectural elements, dependencies, constraints, and internal tensions evolve within the system and are not directly observable from its exterior. Empirical access is therefore mediated through projections such as metrics, events, reports, and aggregated indicators. These projections constitute observable manifestations of enterprise behavior, but they do not establish a one-to-one correspondence with the underlying structural configuration.

This paper addresses the problem of whether, and to what extent, an enterprise can be *structurally diagnosed* on the basis of such observable projections. Structural diagnosability is treated as an epistemic property: it concerns the possibility of inferring structural phase membership from observations, given a fixed formal model and fixed observability conditions. The question is not whether a particular enterprise state is desirable or undesirable, but whether it is distinguishable, in principle, from other structurally different states.

Formally, the enterprise is modeled as a continuum K_{EA} with an internal structural state space and a corresponding observable projection. Distinct internal configurations may map to identical or near-identical observable outcomes. When this occurs, state-level diagnosis is undecidable, even under idealized assumptions of precise and noise-free measurement. This non-uniqueness of the inverse mapping motivates a shift from exact state identification to *phase-level* diagnosability, in which structurally equivalent configurations are grouped into equivalence classes.

The core problem can therefore be stated as follows: given a set of observable manifestations, under what conditions can an enterprise be diagnosed as belonging to a specific structural phase, and under what conditions is such a diagnosis fundamentally underdetermined by the available observations?

Several boundaries are explicit. The paper does not propose mechanisms for improving observability, refining measurement procedures, or influencing enterprise behavior. It introduces no managerial constructs, optimization criteria, or intervention operators. Incomplete or noisy observability is not treated as a practical deficiency to be mitigated, but as a structural condition of the diagnostic setting.

Within these constraints, the objective is to clarify the logical status of diagnostic statements about enterprises. A diagnosis may be possible, impossible, or falsifiable only at the level of structural phases. Making these limits explicit is necessary to distinguish well-founded diagnostic knowledge from claims that exceed what the formal model and observable projections can support.

2 Formal Setting

This section establishes the formal and epistemic context required for the diagnostic analysis developed in this paper. Rather than presenting a complete mathematical model of the enterprise, the formal setting clarifies the logical status of structural claims under conditions of incomplete observability. Its purpose is to delineate the boundaries within which diagnostic statements are meaningful, and to specify the assumptions under which such statements can be assessed as valid, invalid, or underdetermined.

The enterprise is treated as a structured system whose internal configuration is not directly accessible to observation, but only observable through partial and indirect projections. Accordingly, the formal setting does not aim at full state reconstruction, identifiability, or control. Instead, it provides the minimal conceptual and formal apparatus required to reason about distinguishability, equivalence relations, and phase membership in the presence of structural non-identifiability.

A central premise is that non-identifiability is not an artifact of insufficient modeling effort, but an intrinsic property of the mapping between internal structural spaces and observable domains. Distinct internal configurations may induce identical observable projections, even under idealized assumptions of perfect measurement. Under such conditions, inverse inference from observables to unique structural states is not well-defined, and diagnostic claims must be formulated at an appropriate level of abstraction.

As part of this clarification, the present work is explicitly positioned with respect to adjacent literatures. This positioning is not intended as a survey of methods, a comparison of techniques, or an evaluation of empirical performance. Its sole purpose is to identify which commonly adopted assumptions—such as full observability, state identifiability, or controllability—are explicitly not adopted here, and to explain why the failure of identifiability must be treated as a first-class and informative diagnostic outcome rather than as a deficiency to be resolved.

2.1 Positioning with Respect to Related Work

Context and motivation. A broad range of literatures address enterprises and complex socio-technical systems, including systems theory, control and optimization, enterprise architecture (EA) and information systems (IS), and managerial diagnostics. Despite partially overlapping terminology—such as state, structure, observation, and performance—these traditions are typically oriented toward objectives that differ fundamentally from the problem examined here.

The present work focuses on *structural diagnosability* under incomplete observability: whether the available observation surface permits a principled distinction between structurally incompatible enterprise states, or whether such a distinction is formally impossible. The contribution is diagnostic rather than prescriptive. Negative outcomes—non-identifiability, undecidability, or STOP conditions—are treated as informative epistemic results rather than as deficiencies to be remedied.

Systems theory and observability. Classical systems theory studies observability and identifiability for systems whose model class is fixed or parameterized a priori. The central question is whether internal states or parameters can be reconstructed from outputs under known inputs and dynamics, as formalized in both linear and nonlinear settings (e.g. [4, 1, 3]).

The enterprise setting considered here differs in a structurally decisive way. No single fixed model class is assumed: feasible structure is constrained by organizational, technical, legal, and economic boundaries, and these constraints may evolve with the state of the system itself. Observation channels are heterogeneous, partial, and often endogenous. Accordingly, the diagnostic question is not whether a state can be reconstructed within a known model, but whether the available evidence suffices to distinguish among *structurally incompatible* enterprise states. The focus therefore shifts from reconstruction to diagnostic decidability.

Model-based diagnosis and fault isolation. Model-based diagnosis and fault isolation provide formal tools for detecting inconsistencies between observations and a given model, and for isolating fault hypotheses [7, 2]. These approaches presuppose a well-defined fault taxonomy, a stable observation interface, and a clear separation between system structure and control logic.

Enterprises systematically violate these assumptions. Structural incompatibilities are often global rather than local, fault classes are not exhaustively enumerable, and feasibility constraints cannot be factored out of the diagnostic problem. The present work adopts the diagnostic emphasis on explicit hypothesis spaces and admissibility conditions, but without assuming that a complete fault taxonomy or a uniquely identifiable structure exists.

Control theory and optimization. Control and optimization frameworks presuppose a controllable interface, an objective function, and a stable mapping from interventions to outcomes. These assumptions enable performance-oriented reasoning, but they implicitly require that the system state is sufficiently well-defined to support optimization [8].

Structural diagnosability precedes such reasoning. If multiple incompatible enterprise states are consistent with the same observation surface, optimization relative to any single assumed state introduces unwarranted certainty. For this reason, the paper does not propose control policies or optimization schemes. Instead, it isolates a necessary epistemic precondition for rational intervention: that the enterprise state is diagnostically well-posed under available evidence. Where this precondition fails, the correct outcome is a negative diagnostic result.

EA/IS frameworks and architectural practice. EA and IS frameworks provide structured representations of enterprises through layers, viewpoints, and governance processes. Their primary function is to support coordination, communication, and planning. Architectural descriptions are often treated as authoritative abstractions of the underlying enterprise.

From a diagnostic perspective, however, enterprises typically maintain multiple concurrent structural representations (e.g., declared, current, feasible). The mapping from observations to structural claims is neither injective nor stable. In practice, EA resolves this indeterminacy through process, consensus, and social validation. In contrast, the present work treats the issue as one of formal diagnosability: whether the observation surface suffices, in principle, to justify a structural claim independently of governance or agreement mechanisms.

KPI-driven evaluation and managerial case literature. KPI-driven approaches evaluate enterprises through selected indicators, while managerial case literature emphasizes narrative coherence and retrospective causal explanation. Both may support action under uncertainty, but neither directly addresses the epistemic question of whether a structural diagnosis is warranted.

Indicators and narratives can remain actionable even when the underlying structure is not uniquely identifiable. Structural diagnosability, by contrast, requires explicit recognition of non-uniqueness: that the same observations may be compatible with mutually exclusive structural interpretations. Accordingly, this paper avoids KPI logic and case-based prescriptions, and treats non-identifiability as an explicit diagnostic outcome, consistent with classical discussions of falsifiability and underdetermination [5, 6].

What this paper does *not* attempt to do. To avoid scope ambiguity, several exclusions are stated explicitly:

- The paper does not propose an EA method, maturity model, or governance process.
- It does not define KPIs, targets, or performance objectives.
- It does not provide optimization algorithms or intervention policies.
- It does not assume that enterprises can be made fully observable.
- It does not claim superiority in terms of outcomes, efficiency, or effectiveness.

Summary. Relative to adjacent literatures, the distinctive stance of this paper is to treat structural diagnosability under incomplete observability as the primary object of analysis. Negative results are not framed as deficiencies, but as formal boundaries on what can be inferred. In this sense, the contribution is diagnostic-first: it establishes when enterprise states are well-defined objects of inference, and when they are not.

3 Structural diagnosability (epistemic, non-algorithmic)

3.1 Problem statement: what can be known from observations

We consider an enterprise as a continuum K_{EA} with admissible internal state space $\Omega(K_{EA})$ and boundary $\partial\Omega(K_{EA})$, interpreted as a failure or phase-transition boundary. Observations do not provide direct access to $\Omega(K_{EA})$; they arise as manifestations generated through interaction with an environment E and an observation operator Π.

We use the established manifestation map

$$\Sigma_{obs}(K_{EA}, E) := \Pi(\Omega(K_{EA}) \cap \Omega(E)) \,,$$

and interpret any concrete dataset as a finite observation trace $D \subseteq \Sigma_{obs}$ collected under a fixed observational regime Π and environment E.

Diagnosability question. Given a finite observation trace D, what can be inferred about the internal state of K_{EA}—and about its relation to $\partial\Omega(K_{EA})$—without introducing prescriptive assumptions, control intent, or optimization criteria?

The answer is inherently *set-valued*: the available data typically admits multiple internally consistent realizations compatible with the same observations.

3.2 Compatibility set and non-identifiability

Definition 3.1 (Compatibility set)**.** Fix (E, Π) and an observation trace D. The *compatibility set* is defined as the set of all enterprise continua compatible with the observations:

$$\mathcal{K}_{compat}(D; E, \Pi) := \{K_{EA} \ : \ D \subseteq \Sigma_{obs}(K_{EA}, E)\} \,.$$

Definition 3.2 (State-level compatibility (within a fixed K_{EA}))**.** When the formalization distinguishes internal states $x \in \Omega(K_{EA})$, the set of compatible internal states for fixed (K_{EA}, E, Π) is

$$\Omega_{compat}(D \mid K_{EA}, E, \Pi) := \{x \in \Omega(K_{EA}) \ : \ D \subseteq \Sigma_{obs}(x; K_{EA}, E, \Pi)\} \,,$$

where $\Sigma_{obs}(x; \cdot)$ denotes the observable manifestation generated by state x under (E, Π).

Definition 3.3 (Structural diagnosability (epistemic))**.** A diagnostic question Q (e.g. proximity to $\partial\Omega$, or admissibility of a diagnostic regime) is *structurally diagnosable* from D under (E, Π) if Q is invariant over the compatibility set:

$$\forall K_1, K_2 \in \mathcal{K}_{compat}(D; E, \Pi) : \quad Q(K_1) = Q(K_2).$$

For a fixed continuum K_{EA}, an analogous criterion applies at the state level if Q is constant over $\Omega_{compat}(D \mid K_{EA}, E, \Pi)$.

Interpretation. Structural diagnosability is an epistemic property of the triple (D, E, Π) relative to the formal object K_{EA}. It does not assert uniqueness, computability, or algorithmic reconstructability. It specifies the conditions under which a *definite* diagnostic statement is logically justified, and under which it is not.

3.3 Underdetermination as the default

The manifestation map is generally non-injective: distinct internal continua, or distinct internal states within a continuum, may generate indistinguishable observation traces under the same (E, Π).

Generic non-identifiability. Unless the observation operator Π is informationally complete with respect to $\Omega(K_{EA})$—a degenerate condition in enterprise contexts—the mapping $K_{EA} \mapsto \Sigma_{obs}(K_{EA}, E)$ fails to be injective. For finite observation traces D, the compatibility set $\mathcal{K}_{compat}(D; E, \Pi)$ therefore contains multiple elements in typical settings.

Accepted negative result. Reconstruction of the internal structure of an enterprise continuum from observational traces is, in general, impossible. The maximally justified outcome of diagnosis is therefore a *set* of compatible realizations together with statements that are invariant over that set.

3.4 Diagnosable properties: invariants and bounds

Even under non-identifiability, certain structural properties remain diagnosable as invariants or bounds.

Definition 3.4 (Diagnosable invariant)**.** A functional I defined on continua is a *diagnosable invariant* from D if it is constant on $\mathcal{K}_{compat}(D; E, \Pi)$.

Definition 3.5 (Diagnosable bound)**.** A scalar functional B is *diagnosably bounded* from D if there exist functions $b^{-}(D)$ and $b^{+}(D)$ such that for all $K \in \mathcal{K}_{compat}(D; E, \Pi)$,

$$b^{-}(D) \leq B(K) \leq b^{+}(D),$$

with bounds depending only on D under fixed (E, Π).

Examples (structural, non-managerial). Typical candidates include distance-to-boundary proxies, admissibility of transition operators Ψ, or emptiness of the feasible model space $M_{feasible}$. These are statements of structural feasibility, not performance targets or objectives.

3.5 Model-space triad and the STOP boundary

We assume the established triad of model spaces: $M_{current}$ (what is), $M_{declared}$ (what is claimed), and $M_{feasible}$ (what can exist without violating constraints). A central negative diagnostic outcome is the collapse of feasibility.

Definition 3.6 (Feasibility collapse (STOP condition))**.** A trace D implies a diagnostic $STOP$ under (E, Π) if

$$\forall K \in \mathcal{K}_{compat}(D; E, \Pi) : \quad M_{feasible}(K) = \varnothing.$$

Meaning. STOP is a diagnostic classification, not advice. It asserts the non-existence of any internally consistent model compatible with the observations and declared constraints.

3.6 Diagnosability of proximity to $\partial\Omega$

The boundary $\partial\Omega(K_{EA})$ is not directly observable. One may nevertheless ask whether proximity to this boundary is diagnosable as an invariant or bound.

Definition 3.7 (Diagnosable proximity statement)**.** Let $d(\cdot, \partial\Omega)$ be a distance-like functional on internal states or continua. A statement of the form

$$d(\cdot, \partial\Omega) \leq \varepsilon$$

is diagnosable from D if it holds for all compatible realizations.

Accepted limitation. Without strong assumptions on Π and on the metric structure, proximity to $\partial\Omega$ is generally not diagnosable beyond one-sided bounds.

3.7 Epistemic closure: what this section does *not* claim

This section does not claim: (i) unique identification of internal structure; (ii) existence of a computable estimator; (iii) controllability, optimizability, or policy implications.

It establishes the epistemic formalism required to state precisely what is—and what is not—inferable from observations in the K_{EA} setting.

4 Phase logic as diagnostic regimes

4.1 Purpose and stance

Phase logic is introduced as a *diagnostic* language for classifying observed enterprise dynamics into regimes. It is not a control scheme, not a transformation methodology, and not an optimization framework.

A *phase* is a formal diagnostic category: a set of internal realizations (continua or states) sharing invariant properties relevant to feasibility, stability, and admissible transitions under the given formalization.

4.2 Phases as subsets of $\Omega(K_{EA})$ and/or model spaces

Let K_{EA} be fixed. Phases may be defined at different formal levels, depending on which internal structures are made explicit.

Definition 4.1 (State-space phase)**.** A *state-space phase* is a subset $\Phi \subseteq \Omega(K_{EA})$ specified by constraints expressed using existing primitives, such as thresholds Θ, distance to the boundary $\partial\Omega$, admissible flows J, existence of cycles C, continuity indicators k, and feasibility relations.

Definition 4.2 (Model-space phase)**.** A *model-space phase* is a subset

$$\widehat{\Phi} \subseteq \{(M_{current}, M_{declared}, M_{feasible})\}$$

specified by constraints on feasibility, consistency, and compatibility between model spaces.

Observationally induced phase statement. Given observations D collected under (E, Π), phase classification is generally *set-valued* unless diagnosability holds:

$$\Phi_{admissible}(D) := \{\Phi \mid \exists K \in \mathcal{K}_{compat}(D; E, \Pi) \text{ such that } K \in \Phi\}.$$

A unique phase label is justified only if it is invariant over the entire compatibility set.

4.3 Canonical diagnostic regimes (non-prescriptive)

We introduce a minimal, canonical family of diagnostic regimes expressed without adding new primitives. Each regime is defined exclusively through existence or non-existence statements concerning feasibility, continuity, and boundary proximity.

Definition 4.3 (Regime $\mathbf{R_{STOP}}$ (feasibility collapse))**.** $\mathbf{R_{STOP}}$ holds if $M_{feasible} = \varnothing$, as defined in Definition 3.6. This regime represents an epistemic non-existence classification.

Definition 4.4 (Regime $\mathbf{R_{COLLAPSE}}$)**.** $\mathbf{R_{COLLAPSE}}$ holds if the compatibility set implies boundary violation, in the sense that all compatible realizations lie on or beyond the boundary:

$$\forall (K \text{ or } x) \text{ compatible with } D: \quad (K \notin \Omega) \text{ or } x \in \partial\Omega.$$

Equivalently, if a diagnosable bound enforces $d(\cdot, \partial\Omega) \leq 0$.

Definition 4.5 (Regime $\mathbf{R_{INERTIA}}$ (stable but non-expansive))**.** $\mathbf{R_{INERTIA}}$ holds if: (i) feasibility is non-empty, and (ii) under the admissible transition family Ψ, only local transitions are compatible with thresholds, such that no increase in effective structural dimensionality is diagnostically forced.

Definition 4.6 (Regime $\mathbf{R_{ADAPT}}$ (local adaptation))**.** $\mathbf{R_{ADAPT}}$ holds if: (i) feasibility is non-empty, and (ii) at least one compatible realization admits the observed changes as internal reconfiguration within $\Omega(K_{EA})$ without violating thresholds, with continuity preserved (e.g. non-vanishing k over the considered horizon, if defined in the core).

Definition 4.7 (Regime $\mathbf{R_{TRANSITION}}$ (phase transition admissible))**.** $\mathbf{R_{TRANSITION}}$ holds if: (i) feasibility is non-empty, and (ii) at least one compatible realization requires or exhibits a transition across a threshold class that changes admissible structure (e.g. activation of a previously absent axis or cycle family, or a diagnosable change in the admissible set of Ψ-moves), while remaining within $\Omega(K_{EA})$.

Remarks.

- Regimes are defined purely through constraints on existing primitives; no prescriptive interpretation is implied.
- The intended use is diagnostic: to state which regimes are supported, excluded, or undecidable given the evidence.
- Under partial observability, regimes are not necessarily mutually exclusive.

4.4 Phase logic under underdetermination

Because $\mathcal{K}_{compat}(D; E, \Pi)$ is typically non-singleton, phase logic is formulated as a three-valued (or set-valued) diagnostic logic: *ruled out*, *admissible*, or *forced*.

Definition 4.8 (Phase status relative to D)**.** Let $\mathbf{R}$ be any regime predicate. Define:

$$\begin{aligned} \mathbf{R} \text{ ruled out} &\iff \forall K \in \mathcal{K}_{compat}(D; E, \Pi): \ \neg\mathbf{R}(K), \\ \mathbf{R} \text{ forced} &\iff \forall K \in \mathcal{K}_{compat}(D; E, \Pi): \ \mathbf{R}(K), \\ \mathbf{R} \text{ admissible} &\iff \exists K \in \mathcal{K}_{compat}(D; E, \Pi): \ \mathbf{R}(K). \end{aligned}$$

Key consequence. A unique phase label is justified only when some regime is forced and all incompatible regimes are ruled out. Otherwise, the correct diagnostic output is a set of admissible regimes.

Generic non-uniqueness of phase classification. For finite observation traces under partial observability, multiple regimes among $\{\mathbf{R}_{\mathbf{INERTIA}}, \mathbf{R}_{\mathbf{ADAPT}}, \mathbf{R}_{\mathbf{TRANSITION}}\}$ may remain admissible simultaneously. Phase classification is therefore generally not forced.

Interpretation. Ambiguous phase outcomes are not a defect of the framework; they reflect irreducible epistemic limits and prevent unwarranted diagnostic commitment.

4.5 STOP dominance and logical precedence

Some regimes dominate others by logical precedence.

STOP dominance. If $\mathbf{R}_{\mathbf{STOP}}$ is forced by D, then no non-STOP regime can be forced. Any classification asserting a forced non-STOP regime under feasibility collapse is inconsistent with the definition of STOP.

4.6 What phase logic does *not* do

Phase logic does not: (i) recommend actions; (ii) rank alternatives; (iii) encode performance objectives or KPI targets; (iv) assume controllability or optimizability of K_{EA}.

It provides a disciplined language for stating which diagnostic regimes are supported, excluded, or undecidable under explicit non-identifiability.

5 Implications (Epistemic Only)

In this paper, the term "implications" refers exclusively to *epistemic consequences* of the formal setting. These consequences constrain what can be inferred, asserted, or ruled out given a fixed observation interface and a fixed vocabulary boundary (`ETS.K_EA.v1.1`). They do not constitute operational, managerial, normative, or methodological recommendations.

5.1 Implications for Inference Under Incomplete Observability

The formal definitions and results entail the following epistemic limitations:

1. **Non-uniqueness is the default.** In the absence of explicit identifiability conditions, the inverse relation from observations to internal structure is set-valued. Multiple mutually incompatible internal realizations may remain observationally indistinguishable.
2. **Apparent stability may be epistemic rather than structural.** Stable patterns in observations do not entail stability of the underlying internal structure. Such apparent stability may result from observational coarseness, masking effects, or information loss at the observation interface.
3. **Additional data does not guarantee improved diagnosability.** Supplementary observations may be epistemically irrelevant if they do not refine the indistinguishability classes induced by the observation operator. Claims of improved diagnosability therefore require formal justification relative to the model and interface, not intuitive appeal.
4. **Boundary effects dominate near critical regions.** In the vicinity of phase boundaries, including STOP, small changes in observations or assumptions may induce discontinuous changes in admissible inferences. This sensitivity reflects epistemic instability of inference, not necessarily volatility of the underlying system.

5.2 Implications for the Meaning of Phase Regions

A phase-region label derived from the predicates defined in this paper must be interpreted as follows:

- **A classification within the formal semantics.** Phase regions are defined relative to explicit primitives, assumptions, and the declared observation interface. They have no meaning outside this formal context.
- **Not an assessment of desirability.** A region is neither favorable nor unfavorable in a managerial, operational, or normative sense. It expresses only diagnosability and existence conditions within the model.
- **Not a guarantee of recoverability or control.** Membership in a non-STOP region does not imply that the system is recoverable, controllable, stabilizable, or improvable under any intervention scheme.

5.3 Implications of STOP

The STOP classification has the following epistemic implications:

1. **Inference collapse.** Under the given observation interface and formal assumptions, non-degenerate structural diagnosis is not epistemically justified.
2. **Constraint violation or undecidability.** STOP indicates either violation of existence or consistency constraints, or a loss of diagnosability required to support meaningful structural claims within the admitted vocabulary.

3. **Non-substitutability.** STOP cannot be replaced by any positive diagnostic classification without introducing additional assumptions, new primitives, or semantic extensions beyond the declared formal boundary.

5.4 Implications for Claims and Reporting

The formal results impose negative constraints on what can be claimed in a scientifically valid report:

1. **No causal claims without identifiability.** When multiple incompatible internal realizations are compatible with the same observations, causal assertions about internal structure are epistemically underdetermined.
2. **No operational conclusions from epistemic classifications.** Translating a diagnostic classification into a decision, recommendation, or intervention requires an explicit policy, control, or optimization model, none of which is provided or assumed here.
3. **No universality without a declared mapping.** Any application to a concrete enterprise must explicitly specify the mapping from empirical artifacts and observations to the formal objects of the model. Without such a mapping, reported results lack determinate interpretive content.

6 Limits and Non-Claims

This document presents a strictly *diagnostic* formalization of (i) structural diagnosability conditions and (ii) phase-logic predicates, including STOP, under the assumptions and symbol vocabulary fixed by the formal setting and by `ETS.K_EA.v1.1`. It does not provide prescriptions, control rules, optimization objectives, or decision guidance.

6.1 Non-Claims (Explicit)

The following statements are explicitly *not* asserted anywhere in this paper:

1. **No control or optimization claim.** The paper does not claim the ability to control, optimize, stabilize, or improve an enterprise system, nor to select actions that move the system within or between phase regions.
2. **No algorithmic or procedural claim.** The paper does not provide algorithms, procedures, heuristics, or operational recipes for computing diagnoses or enforcing constraints in real organizations.
3. **No KPI or managerial metric claim.** The paper does not define KPIs, performance targets, maturity models, or managerial scorecards. Any numerical illustration, if present, serves only as an internal demonstration of formal predicates.
4. **No causal attribution from symptoms to structure.** The paper does not claim that observed incidents or measurements uniquely identify internal structure. In general, multiple distinct internal configurations may be compatible with the same observable manifestations.
5. **No completeness claim under partial observability.** Under incomplete observability, the framework does not guarantee that every true internal failure mode or degradation is diagnosable from the available observations.

6. **No universality beyond the formal setting.** The definitions are not claimed to be universally adequate for every organization without an explicit, justified mapping from that organization to the formal objects constrained by `ETS.K_EA.v1.1`.

6.2 Why Control and Optimization Questions Are Ill-Posed Here

A recurrent misreading is to treat any phase-space representation as implicitly actionable. Within the present formalism, such an inference is invalid for structural reasons:

1. **Diagnostic statements are not policies.** A diagnostic statement has the form: "given assumptions A and observations O, the set of compatible internal realizations is $\mathcal{K}_{\mathrm{compat}}$". A control or optimization statement requires additional objects: a policy or action space, a transition model under interventions, and a goal functional. None of these objects are introduced here, and introducing them would violate the diagnostic-only scope.
2. **Underdetermination under partial observability.** When observations are incomplete, inverse inference is generally non-unique. Without uniqueness, optimization of interventions is not well-defined, since different compatible internal realizations may require incompatible actions.
3. **Specification boundary (no new primitives).** `ETS.K_EA.v1.1` fixes the admissible vocabulary and precludes the introduction of managerial primitives such as objectives, utilities, or ROI. Questions of the form "what should be done?" therefore lie outside the formal language of this paper.
4. **Category distinction: epistemic versus operational.** The results constrain what can be *known*, *ruled out*, or *left undecided* from a given observation interface. They do not specify what can be *implemented* in a socio-technical organization.

Accordingly, control and optimization questions are not merely unanswered; they are *ill-posed within the present formalism.*

6.3 STOP as a Valid Scientific Outcome

STOP is treated as a first-class outcome because it can be the correct diagnostic conclusion under the formal semantics:

- **STOP is an epistemic classification.** STOP indicates that, under the admitted predicates and available observations, diagnosability fails, existence or consistency constraints are violated, or the formal interface does not support non-degenerate inference.
- **STOP may be maximally informative.** In certain regimes, the statement "no non-degenerate diagnosis is justified within this interface" is strictly more informative than any positive but unsupported classification.
- **STOP is falsifiable at the interface level.** If an observational regime classified as STOP nevertheless permits stable and unique inference under the same formal assumptions, then the STOP predicate, or its prerequisites, is incorrect and must be revised *within the admitted vocabulary.*

6.4 Falsifiability Boundaries

The paper contains two classes of claims, each with a distinct falsifiability mode:

(F1) Internal formal claims. Statements of the form "given Definitions X and Assumptions Y, Proposition Z holds" are falsified by a *formal counterexample*: an explicit construction within the formal setting that satisfies X and Y while violating Z.

(F2) Projection and interface claims. Statements relating formal objects to an empirical enterprise are conditional on a declared observational interface. They are falsified if at least one of the following can be demonstrated:

1. **Specification violation:** the argument requires primitives not permitted by `ETS.K_EA.v1.1`, or contradicts an explicit constraint of that specification.
2. **Incoherent mapping:** the observational interface cannot be instantiated without altering the formal meaning of the symbols, i.e., the mapping is not semantics-preserving.
3. **Interface insufficiency:** for a claimed diagnosability regime, the observation channel is provably insufficient, for example because information loss renders the inverse problem non-identifiable under the stated assumptions.
4. **Indeterminate predicate evaluation:** under the same formal assumptions and the same declared interface, the diagnostic predicates cannot be evaluated unambiguously.

6.5 Falsifiability by Counterexample Templates

The paper additionally provides explicit counterexample templates. These specify what would count as a counterexample *in principle* under admissible assumptions, without prescribing any operational search procedure or empirical protocol.

6.6 Falsifiability by example: counterexample templates (epistemic)

This subsection provides *counterexample templates* in a strictly epistemic sense. Each template specifies: (i) admissible assumptions, (ii) an existential construction class, and (iii) the proposition type that would be refuted *if* such a construction exists. No operational testing, algorithmic procedure, or empirical search protocol is implied.

CE-1 (Non-separability under the adopted observational interface).

1. **Assumptions.** Fix the observational regime (E, Π) and the induced notion of admissible observation traces $D \subseteq \Sigma_{obs}(K_{EA}, E)$ as defined in Section 3. Assume the formal setting permits reasoning over the compatibility set $\mathcal{K}_{compat}(D; E, \Pi)$.
2. **Construction class.** Two distinct internal structures (two distinct continua $K_{EA}^{(1)} \neq K_{EA}^{(2)}$ admissible under the formal setting) such that the *same* observation trace is admissible for both:

$$D \subseteq \Sigma_{obs}(K_{EA}^{(1)}, E) \quad \text{and} \quad D \subseteq \Sigma_{obs}(K_{EA}^{(2)}, E).$$

   Equivalently, an existential witness that $|\mathcal{K}_{compat}(D; E, \Pi)| > 1$ holds for the claimed diagnosability regime.

3. **Falsified proposition (if such constructions exist).** Any proposition asserting *uniqueness* of structural inference from D under the stated (E, Π) (i.e., $|\mathcal{K}_{compat}(D; E, \Pi)| = 1$ or an equivalent uniqueness claim) is refuted by a single CE-1 witness.

CE-2 (STOP false positive under admitted realizations).

1. **Assumptions.** A STOP predicate is used as an epistemic classifier and is intended to hold only when diagnosis degenerates under the formal semantics and admissibility assumptions of the paper.
2. **Construction class.** An admissible situation in which STOP is asserted, yet there exists at least one compatible realization supporting non-degenerate epistemic diagnosis under the same formal setting:

   $$\mathrm{STOP}(D; E, \Pi) = \text{true} \quad \text{and} \quad \exists K \in \mathcal{K}_{compat}(D; E, \Pi) \text{ such that the claimed diagnostic statement is w}$$

   The construction is purely existential; no algorithmic realization is implied.
3. **Falsified proposition (if such constructions exist).** Any proposition asserting *soundness* of the STOP predicate in the sense "STOP implies diagnosability loss under the formal semantics" is refuted.

CE-3 (STOP false negative under diagnosability loss).

1. **Assumptions.** The formal scope includes an admissible state space and a notion of diagnosability loss (e.g., feasibility collapse or non-identifiability to the point that no invariant diagnostic statement remains).
2. **Construction class.** A situation in which STOP is *not* asserted, yet diagnosability has degenerated under the same formal assumptions:

   $$\mathrm{STOP}(D; E, \Pi) = \text{false} \quad \text{and} \quad \forall \text{ diagnostic statements } Q \text{ in scope, } Q \text{ fails to be invariant over } \mathcal{K}_{compa}$$

   Equivalently, the formal prerequisites for meaningful diagnosis fail while STOP remains false.
3. **Falsified proposition (if such constructions exist).** Any proposition asserting *completeness* of the STOP predicate with respect to the stated notion of diagnosability loss is refuted.

CE-4 (Phase assignment non-invariance under admissible equivalences).

1. **Assumptions.** Phase logic assigns phase regions as epistemic classifications intended to be invariant under admissible equivalences (e.g., relabelings or semantics-preserving encodings) within the formal setting.
2. **Construction class.** Two representations of the *same* admitted underlying structure (equivalent in the intended formal sense) that nevertheless yield different phase assignments:

   $$K_{EA}^{(1)} \equiv K_{EA}^{(2)} \quad \text{but} \quad \mathrm{Phase}(K_{EA}^{(1)}; E, \Pi) \neq \mathrm{Phase}(K_{EA}^{(2)}; E, \Pi).$$

3. **Falsified proposition (if such constructions exist).** Any proposition claiming that

phase labels are well-defined on equivalence classes (i.e., representation-invariant) is refuted by a single CE-4 witness.

Remark (status of falsification). These templates define *logical refuters* for proposition types used in the paper. They do not prescribe how to find such constructions, and they make no claim about empirical prevalence. The falsification mode is epistemic: existence of a single admissible counterexample suffices.

6.7 Hard Limits of the Framework

Even within its intended scope, the framework cannot guarantee:

1. **Uniqueness of inferred internal structure** from observations, unless explicit identifiability conditions are met.
2. **Completeness of diagnosis** under partial observability; some internal degradations may remain observationally indistinguishable.
3. **Operational feasibility of interventions,** since interventions are not elements of the formal language employed here.
4. **Empirical validity of a chosen mapping** from a real organization to the formal objects; such mappings require external evidence and validation beyond the scope of this paper.

6.8 Interpretive Takeaway

The correct reading is:

> This paper specifies the conditions under which structural diagnosis is well-posed in a formal sense, and the conditions under which the only honest diagnostic outcome is STOP. It does not prescribe actions, objectives, or optimization criteria.

7 Conclusion

This paper has developed a strictly diagnostic framing of *structural diagnosability* for enterprise-scale systems under conditions of incomplete observability. Its central contribution is not the proposal of a control mechanism, optimization strategy, or governance prescription, but a formal clarification of when diagnostic statements are epistemically well-posed, and when they are not.

The first contribution is the formulation of diagnosability as an epistemic property of the observable projection, rather than as a function of intervention capability, data volume, or analytical effort. By explicitly separating observable manifestations from internal structure, the paper defines diagnosability in terms of whether the available observations support a non-empty and non-degenerate set of internally consistent structural hypotheses. Loss of diagnosability is thus characterized as a structural condition imposed by the observation interface and formal constraints, not as a deficiency of measurement practice or methodological sophistication.

The second contribution is the introduction of a phase-based logic of diagnosability. Instead of treating diagnosability as a continuous or monotonic quantity, the paper formalizes discrete diagnostic regimes and the threshold relations between them. Within this logic, phase transitions correspond to qualitative changes in the space of admissible structural hypotheses. They do not represent managerial choices, policy shifts, or intervention decisions, but changes in what can be meaningfully inferred under fixed formal assumptions.

The third contribution is the elevation of *STOP* to a first-class diagnostic outcome. STOP is not interpreted as inaction, failure, or recommendation, but as a formally valid conclusion reached when diagnosability collapses under the adopted semantics. In such situations, any prescriptive or optimization-oriented claim would be epistemically unjustified. Treating STOP as a legitimate terminal diagnostic result is therefore necessary for maintaining scientific coherence in regimes of severe underdetermination.

The falsifiability boundaries of the framework are made explicit. The core claims would be refuted if it were shown that: (a) diagnosability can be preserved despite an empty or internally inconsistent compatibility set; (b) discrete diagnostic phase distinctions can be eliminated in favor of a fully continuous regime without loss of explanatory adequacy; or (c) terminal STOP conditions can be replaced by universally valid prescriptive rules without introducing additional structural assumptions or primitives. Any such result would contradict the formal phase logic and boundary conditions established here.

Equally explicit are the non-claims. This paper does not propose methods for controlling, optimizing, steering, or stabilizing enterprises. It does not provide decision rules, best practices, governance models, or performance improvements. No implication is made that diagnosability entails controllability, or that correct diagnosis enables successful intervention. All prescriptive interpretations lie strictly outside the scope of the present formalism.

The present analysis depends on earlier formal artifacts for its definitions and primitives. In particular, the ontological grounding and operator semantics are inherited from the archival artifacts commonly referred to as PDF-0 and PDF-1. The current paper should therefore be read as a dependent, non-expansive contribution within that artifact lineage, introducing no new primitives or semantic extensions.

Any future work, if undertaken, is limited to artifact-level refinement rather than conceptual expansion. This includes consolidation of references, tightening of selected proof sketches, and the possible inclusion of executable or mechanically checkable demonstrations within the existing formal framework (e.g. at the level of PDF-1–class artifacts). No new diagnostic claims, operational guidance, or prescriptive extensions are implied by these directions.

In summary, this paper establishes diagnosability as a structurally bounded epistemic property, introduces a phase logic for reasoning about its loss, and legitimizes STOP as a scientifically necessary outcome. Its contribution lies in clarifying formal limits rather than proposing solutions, and in enforcing epistemic discipline where interventionist interpretations are not supported by the available structure.

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